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# Capstone Project: Online Course Enrollment and Progress Tracker

**Objective:**
Build a system to track which students enroll in which courses, monitor their progress, and generate basic reports such as completion rates, dropouts, and popular courses.

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## Week 1 – Database Foundations: MySQL & MongoDB

### Tools: MySQL, MongoDB

**Capstone Tasks:**
- Create MySQL tables: `students`, `courses`, `enrollments`, `progress`
- Perform CRUD operations (e.g., new enrollment, update progress)
- Write a stored procedure to calculate course completion percentage for a student
- Use MongoDB to store optional feedback or reviews (JSON)
- Add indexes to search by student or course quickly

**Deliverables:**
- SQL script with schema, CRUD, and stored procedure
- MongoDB script with sample feedback and indexing

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## Week 2 – Data Preprocessing in Python

### Tools: Python (Pandas, NumPy)

**Capstone Tasks:**
- Load enrollment and progress data from CSV or mock API
- Clean invalid entries (missing names, dates, percentages)
- Use `numpy` to calculate average progress
- Use `pandas` to group by course and compute average completion rates

**Sample Code Snippet:** ```python import pandas as pd import numpy as np

df = pd.read_csv("progress.csv") df['completion'] = df['completion'].fillna(0) df['completion'] = np.clip(df['completion'], 0, 100)

summary = df.groupby('course_id')['completion'].mean() print(summary) ```

**Deliverables:**
- Cleaned data showing student progress
- Python report of course-level averages

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# Week 3 – PySpark for Course Analysis

### Tools: PySpark

**Capstone Tasks:**
- Load large enrollment and progress data using PySpark
- Join tables to get course-wise progress
- Group by course to count total enrolled and completed students

**Deliverables:**
- PySpark script with joins and group aggregations
- Output showing top completed and dropped-out courses

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# Week 4 – ETL in Azure Databricks

### Tools: Azure Databricks

**Capstone Tasks:**
- Load cleaned data into Databricks
- Join student and course data
- Create a final table with: student name, course, enrollment date, progress
- Save results in Delta or CSV for dashboarding

**Deliverables:**
- Databricks notebook
- Final clean output table (Delta/CSV)

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# Week 5 – Pipeline Automation with Azure DevOps

### Tools: Azure DevOps

**Capstone Tasks:**
- Automate weekly reporting pipeline
- Run script to generate list of students with less than 50% progress
- Output report/log for follow-up

**Deliverables:**
- Azure DevOps YAML pipeline
- Weekly progress report output

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## Final Outcome by Week 5:

- Structured system to track course enrollments and student progress
- Cleaned data ready for analysis and reporting
- Aggregations for course popularity and completion
- Simple ETL with Databricks
- Automated insights via Azure DevOps pipeline

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